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Unimodality of the independence polynomial of clique corona graphs*

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Abstract

For a positive integer p, a graph G belongs to the class $\mathbf{W}_p$ if $|V(G)| \geq p$ and any p disjoint independent sets can be extended to p disjoint maximum independent sets. In this paper, we establish several inequalities for the coefficients of the independence polynomial of graphs in $\mathbf{W}_p$. We then prove that the clique corona graph $G \circ K_p$, formed from a graph G and a complete graph K_p, also belongs to $\mathbf{W}_p$. As an application, we examine the unimodality problem for the independence polynomial of $G \circ K_p$. In particular, we prove that the independence polynomial of $S_n \circ K_p$, where S_n is a star graph, is unimodal.

Keywords: Independence polynomial, unimodality, clique corona graph, well-covered graph.

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1 Introduction

Throughout this paper, by a graph we mean a finite simple graph without isolated vertices. For a graph G, let $V(G)$ denote the vertex set and $E(G)$ denote the edge set of G, and $n(G) = |V(G)|$. An edge $e \in E(G)$ connecting two vertices x and y will also be written as xy. In this case, it is said that x and y are *adjacent*. The neighborhood of a vertex x in G is the set $N(x) = \{y \in V(G) \mid xy \in E(G)\}$ and the closed neighborhood of x in G is the set $N[x] = \{x\} \cup N(x)$. For a subset S of $V(G)$, let $G[S]$ be the induced subgraph of G on the vertex set S, and we use $G - S$ to denote $G[V(G) - S]$. If $S = \{x\}$, we write $G - x$ instead of $G - \{x\}$. An *independent* (or *stable*) *set* in G is a set of pairwise non-adjacent vertices. An independent set of G is *maximal* if it cannot be extended to a larger independent set. An independent set of the largest size is a *maximum independent set*; its cardinality is denoted $\alpha(G)$ and called the independence number (or stability number) of G.

A *matching* in a graph G is a set of pairwise non-incident edges. Let A and B be two subsets of $V(G)$. A is *matched into* B if there exists a matching in the graph whose set of A-endpoints is exactly A and whose set of opposite endpoints is a subset of B. In 1982, Berge established the following useful characterization of maximum independent sets.

Theorem 1.1 ([3])**.** *In a graph G, an independent set S is maximum if and only if every independent set disjoint from S can be matched into S.*

A graph is *well–covered* if all its maximal independent sets are of the same size [21]. For example, every complete graph on n vertices, denoted by K_n, is well-covered. The only cycles that are well-covered are those with $n = 3, 4, 5$, or 7. Moreover, the path on n vertices, denoted by P_n, is well-covered if and only if $n = 1, 2$, or 4. To classify well-covered graphs, Staples [24] introduced the following family of graphs. For any positive integer p, a graph G belongs to the class $\mathbf{W}_p$ if $n(G) \geq p$ and every p pairwise disjoint independent sets in G are included in p pairwise disjoint maximum independent sets. Clearly, with the assumption that $G \in \mathbf{W}_p$, it implies $n(G) \geq p \cdot \alpha(G)$, $K_p \in \mathbf{W}_p$, for every p, and $\mathbf{W}_1 \supseteq \mathbf{W}_2 \supseteq \mathbf{W}_3 \supseteq \cdots$, where $\mathbf{W}_1$ is the family of all well-covered graphs. Several ways to build graphs belonging to the class $\mathbf{W}_p$ are presented in [20, 24].

If $s_k = s_k(G)$ denotes the number of independent sets of cardinality k in the graph G, then

$$I(G; x) = \sum_{k=0}^{\alpha(G)} s_k x^k = s_0 + s_1 x + \cdots + s_{\alpha(G)} x^{\alpha(G)}$$

is called the *independence polynomial* of G. The independence polynomial, first introduced by Gutman and Harry in 1983 [8], has been widely studied in several articles, in which unimodality problems are extensively investigated; see [22] for further details.

A finite sequence of real numbers $(a_0, a_1, a_2, \ldots, a_n)$ is said to be *unimodal* if there is some $k \in \{0, 1, \ldots, n\}$, called the *mode* of the sequence, such that

$$a_0 \leq \cdots \leq a_{k-1} \leq a_k \geq a_{k+1} \geq \cdots \geq a_n.$$

A polynomial is called *unimodal* if the sequence of its coefficients is unimodal. Alavi, Malde, Schwenk and Erdős proved that for any permutation π of $\{1, 2, \ldots, \alpha\}$ there is a graph G with $\alpha(G) = \alpha$ such that $s_{\pi(1)} < s_{\pi(2)} < \cdots < s_{\pi(\alpha)}$ [1]. Nevertheless, for trees, they stated the following conjecture, which is still open.

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Conjecture 1.2 ([1]). *The independence polynomial of a tree is unimodal.*

A considerable amount of literature has been published on the unimodality of the independence polynomial of a graph. For instance, a famous result of Chudnovsky and Seymour in [4] states that all the roots of $I(G;x)$ are real, whenever G is a claw-free graph, which also proves the unimodality of $I(G;x)$ for all claw-free graphs G.

Let $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ be graphs with disjoint vertex sets. The disjoint union G_1 and G_2, denoted by $G_1 \cup G_2$, is the graph with the vertex set $V_1 \cup V_2$ and the edge set $E_1 \cup E_2$. In particular, kG denotes the disjoint union of $k > 1$ copies of the graph G. The *join*, also known as *Zykov's sum*, of two graphs G_1 and G_2, is the graph $G_1 + G_2$ with vertex set $V(G_1) \cup V(G_2)$, and edge set

$$E(G_1) \cup E(G_2) \cup \{xy \mid x \in V(G_1), y \in V(G_2)\},$$

which includes all edges of G_1 and G_2 together with all edges connecting every vertex of G_1 to every vertex of G_2.

We recall the definition of the corona of graphs, which will be used in the sequel. Let $\mathcal{H} = \{H_v : v \in V(G)\}$ be a family of nonempty graphs indexed by the vertex set of a graph G. *The corona* $G \circ \mathcal{H}$ of G and $\mathcal{H}$ is defined as the disjoint union of G and all graphs H_v, $v \in V(G)$, with additional edges joining each vertex $v \in V(G)$ to all the vertices of H_v [6]. In particular, when all graphs H_v in $\mathcal{H}$ are isomorphic to a common graph H, we simply write $G \circ H$ instead of $G \circ \mathcal{H}$. Furthermore, if all graphs H_v in $\mathcal{H}$ are complete graphs, then $G \circ \mathcal{H}$ is called *clique corona graph*.

Gutman formulated several methods to compute the independence polynomial by utilizing various graph operations.

Lemma 1.3 ([7, 8]). *Let G_1 and G_2 be graphs with disjoint vertex sets. The following assertions are true:*

(i) $I(G_1 \cup G_2; x) = I(G_1; x) \cdot I(G_2; x)$;

(ii) $I(G_1 \circ G_2;\, x) = (I(G_2;\, x))^n \cdot I\left(G_1; \frac{x}{I(G_2;\, x)}\right)$, *where* $n(G_1) = n$.

The following equality is very useful for calculating the independence polynomial.

Lemma 1.4 ([9]). *Let G be a graph, and $v \in V(G)$. Then the following equality holds:*

$$I(G;x) = I(G - v; x) + x \cdot I(G - N[v]; x).$$

For well-covered graphs, Michael and Traves demonstrated that the sequence of coefficients of the independence polynomial is partially increasing, as follows:

Proposition 1.5 ([19]). *If G is a well-covered graph and $I(G;x) = \sum_{k=0}^{\alpha(G)} s_k x^k$, then $s_{k-1} \leq s_k$ for each $1 \leq k \leq \frac{\alpha(G)+1}{2}$.*

Inspired by the above result on the partial increasing property of the coefficients of the independence polynomial of well-covered graphs, we continue to investigate the partial unimodality of graphs in the class $\mathbf{W}_p$ for $p \geq 2$. To establish precise relationships between consecutive coefficients of the independence polynomial of $\mathbf{W}_p$ graphs, it is necessary to study their structure through a special class of graphs called λ-quasi-regularizable.

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Specifically, a graph G is *λ-quasi-regularizable* if $\lambda > 0$ and $\lambda \cdot |S| \leq |N(S)|$ is true for every independent set S of G [16]. Clearly, λ-quasi-regularizable graphs have no isolated vertices. If $\lambda = 1$, then G is a *quasi-regularizable* graph [3]. It is known that well-covered graphs without isolated vertices are quasi-regularizable [3].

Theorem 1.6 ([16]). *Let G be a λ-quasi-regularizable graph. Then the following assertions are true:*

(i) $(k+1) \cdot s_{k+1} \leq (n(G) - (\lambda+1)k) \cdot s_k, 0 \leq k < \alpha(G)$.

(ii) *If* $n(G) \leq (\lambda+2)\alpha(G) - (\lambda+1)$, *then* $s_r \geq s_{r+1} \geq \cdots \geq s_\alpha$, *where* $r = \left\lceil \frac{n(G)-1}{\lambda+2} \right\rceil$.

The paper is organized as follows. In Section 2, we provide some preliminaries on independence polynomials of graphs belonging to the class $\mathbf{W}_p$. In Section 3, we determine the independence polynomial of the clique corona graph $G \circ \mathcal{H}$ and derive an explicit deleted-subgraph formula for its coefficients, together with several low-degree expansions in the non-uniform case. In the uniform case, this yields a closed expression for $I(G \circ K_p; x)$ in terms of the coefficients of $I(G; x)$. In Section 4, we discuss the unimodality of $I(G \circ K_p; x)$ in Theorem 4.6, while Corollary 4.7 specifies the corresponding results concerning $I(G \circ K_1; x)$. As an application, in Section 5, we demonstrate the unimodality of independence polynomials of clique corona graphs $G \circ K_p$, where $G = S_n$ – a star graph on $n+1$ vertices. In the Conclusions section, we present several open problems and potential directions for future research, which are motivated by the findings and discussions in the paper.

2 Some preliminary properties of class $\mathbf{W}_p$

In this section, we recall some known results and establish a relationship for the coefficients of the independence polynomial of graphs in the class $\mathbf{W}_p$.

Theorem 2.1 ([23]). *Let G_0 be an arbitrary graph with $V(G_0) = \{v_1, v_2, \ldots, v_n\}$, and let $G_1, G_2, \ldots, G_n$ be graphs in the class $\mathbf{W}_p$ with $p \geq 2$. Construct a new graph G by identifying one point from each G_i with the corresponding v_i of G_0 for $i = 1, 2, \ldots, n$. Then G belongs to the class $\mathbf{W}_{p-1}$ with $\alpha(G) = \sum_{i=1}^{n} \alpha(G_i)$.*

Let $G_1, G_2, \ldots, G_n$ be complete graphs of orders $p_1, p_2, \ldots, p_n$, respectively, with $\min_{1 \leq i \leq n} p_i \geq 2$. Then, the identifying operation described in Theorem 2.1 is equivalent to building the corona of G with complete graphs of orders $p_1 - 1, \ldots, p_n - 1$.

Corollary 2.2. *Let G be a graph, and let $\mathcal{H} = \{H_v : v \in V(G)\}$ be a family of complete graphs K_{p_v}. Then the clique corona graph $G \circ \mathcal{H}$ belongs to the class $\mathbf{W}_p$, where $p = \min_{v \in V(G)} p_v$ and its independence number satisfies $\alpha(G \circ \mathcal{H}) = n(G)$.*

Note that the above result, in the case $p = 1$, was established in [25]. Furthermore, a characterization of the clique corona graphs for which $G \circ K_p$ belongs to $\mathbf{W}_2$ was also provided in [17]. For example, all the graphs in Figure 1 are of the form $G \circ \mathcal{H}$, but only G_1 is not well-covered, while G_3 is in $\mathbf{W}_2$.

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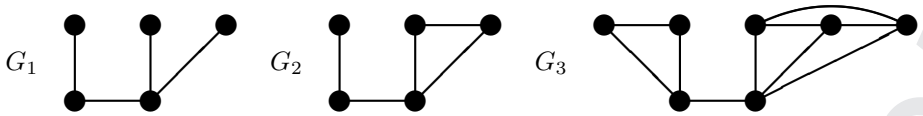


Figure 1: $G_1 = P_2 \circ \{K_1, 2K_1\}$, $G_2 = P_2 \circ \{K_1, K_2\}$, $G_3 = P_2 \circ \{K_2, K_3\}$.

Corollary 2.3. *The graph $G \circ K_p$ belongs to the class $\mathbf{W}_p$ and $\alpha(G \circ K_p) = n(G)$.*

Let $\mathrm{Ind}(G)$ denote the collection of all independent sets in G, while $\Omega(G)$ be the collection of all maximum independent sets in G. Building on previous work, the following result generalizes a corresponding finding for $\mathbf{W}_2$ presented in [18] to the class $\mathbf{W}_p$ with $p \geq 3$.

Theorem 2.4. *Let G be a graph in $\mathbf{W}_p$ with no isolated vertices, and let A be a non-maximum independent set in G. Then the following assertions hold:*

(i) *There are at least p pairwise disjoint independent sets $B_1, B_2, \ldots, B_p$ such that $A \cup B_i \in \Omega(G)$ and $A \cap B_i = \emptyset$ for each $i = 1, 2, \ldots, p$.*

(ii) *If $p \geq 2$, then there are at least $p-1$ pairwise disjoint maximum independent sets $S_1, S_2, \ldots, S_{p-1}$ such that $A \cap S_i = \emptyset$ for each $i = 1, 2, \ldots, p-1$.*

Proof. (i) We proceed by induction on p.

The case $p = 1$ is clear, since $G \in \mathbf{W}_1$, i.e., G is well-covered, and, by definition, there exists an independent set, say B, such that $A \cup B \in \Omega(G)$ and $A \cap B = \emptyset$.

Assume that $G \in \mathbf{W}_p$ with $p \geq 2$. Hence, $G \in \mathbf{W}_{p-1}$, because $\mathbf{W}_p \subseteq \mathbf{W}_{p-1}$. By the induction hypothesis, there exist $p-1$ pairwise disjoint independent sets $B_1, B_2, \ldots, B_{p-1}$ such that $A \cup B_i \in \Omega(G)$ and $A \cap B_i = \emptyset$ for every $i \in \{1, 2, \ldots, p-1\}$. Since $G \in \mathbf{W}_p$, there are p pairwise disjoint maximum independent sets $S_1, S_2, \ldots, S_p$ such that $B_i \subset S_i$ for every $i \in \{1, 2, \ldots, p-1\}$ and $A \subset S_p$. Let $B_p = S_p - A$. Thus we have p pairwise disjoint independent sets $B_1, B_2, \ldots, B_p$ such that $A \cup B_i \in \Omega(G)$ and $A \cap B_i = \emptyset$ for every $i = 1, 2, \ldots, p$.

(ii) Let us choose $p-1$ different vertices $v_1, v_2, \ldots, v_{p-1}$ not belonging to A. In accordance with the definition of the class $\mathbf{W}_p$, one can enlarge the p independent sets $\{v_1\}, \{v_2\}, \ldots, \{v_{p-1}\}, A$ to p pairwise disjoint maximum independent sets $S_1, S_2, \ldots, S_p$ such that $\{v_i\} \subset S_i$ for every $i \in \{1, 2, \ldots, p-1\}$ and $A \subset S_p$. Thus $S_1, S_2, \ldots, S_{p-1}$ are $p-1$ maximum independent sets disjoint from A we were looking for. □

Theorem 2.4 and Corollary 2.3 immediately imply the following.

Corollary 2.5. *If $H = G \circ K_p$ with $p \geq 2$, then for each non-maximum independent set A in H there are at least p pairwise disjoint independent sets $B_1, B_2, \ldots, B_p$ such that $A \cup B_i \in \Omega(H)$ and $A \cap B_i = \emptyset$ for each $i = 1, 2, \ldots, p$.*

Denote $\Omega_k(G) = \{S \in \mathrm{Ind}(G) : |S| = k\}$, where $1 \leq k \leq \alpha(G)$. For $1 \leq k < \alpha(G)$, we define the bipartite graph $H_k(G) = (\Omega_k(G), \Omega_{k+1}(G), \mathcal{E}_k)$, where the bipartition is $\{\Omega_k(G), \Omega_{k+1}(G)\}$ and an edge $SU \in \mathcal{E}_k$ exists if and only if $S \subset U$. This construction provides a convenient framework for studying inclusion relations between independent sets of different sizes. Clearly, the cardinality of Ω_k is s_k. Based on the above, we obtain the following.

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Theorem 2.6. *Let G be a connected graph in the class $\mathbf{W}_p$ of order $n > p$, and let $\alpha = \alpha(G)$. Then the following assertions are true:*

(i) $p \cdot (\alpha - k) \cdot s_k \leq (k+1) \cdot s_{k+1}$, $0 \leq k < \alpha$;

(ii) $s_0 \leq s_1 \leq \cdots \leq s_{\lfloor p \cdot \frac{\alpha+1}{p+1} \rfloor}$;

(iii) G *is* $(p-1)$*-quasi-regularizable;*

(iv) $(k+1) \cdot s_{k+1} \leq (n - p \cdot k) \cdot s_k$, $0 \leq k < \alpha$;

(v) *If* $n \leq (p+1) \cdot \alpha - p$, *then* $s_{\lceil \frac{n-1}{p+1} \rceil} \geq s_{\lceil \frac{n-1}{p+1} \rceil + 1} \geq \cdots \geq s_\alpha$.

Proof. (i) Since G is a connected graph in $\mathbf{W}_p$, we can infer that $p \cdot \alpha \cdot s_0 = p \cdot \alpha \leq s_1 = n$.

Every $U \in \Omega_{k+1}(G)$ has $k+1$ subsets in $\Omega_k(G)$, which means that the degree of every vertex U in H_k is equal to $k+1$. Consequently, $|\mathcal{E}_k| = (k+1) \cdot |\Omega_{k+1}(G)| = (k+1) \cdot s_{k+1}$. On the other hand, by Theorem 2.4 (i), every $S \in \Omega_k(G)$ can be extended by p pairwise disjoint independent sets $B_1, B_2, \ldots, B_p$ such that $S \cup B_1, S \cup B_2, \ldots, S \cup B_p$ are maximum independent sets in G. In other words, the degree of every vertex $S \in \Omega_k(G)$ is at least $p \cdot (\alpha - k)$.

In conclusion, we obtain $p \cdot (\alpha - k) \cdot s_k \leq (k+1) \cdot s_{k+1}$.

(ii) According to Part (i), we have

$$s_k \leq \frac{k+1}{p \cdot (\alpha - k)} \cdot s_{k+1},$$

which ensures that $s_k \leq s_{k+1}$ for every k satisfying $\frac{k+1}{p \cdot (\alpha - k)} \leq 1$, i.e., for $k \leq \frac{p \cdot \alpha - 1}{p+1}$, at least. In other words, the monotone part of the sequence of coefficients goes up to

$$k+1 \leq \left\lfloor \frac{p \cdot \alpha + p}{p+1} \right\rfloor = \left\lceil \frac{p \cdot \alpha}{p+1} \right\rceil.$$

Indeed, if $\frac{p\alpha}{p+1} \in \mathbb{Z}$ then

$$\left\lfloor \frac{p \cdot \alpha + p}{p+1} \right\rfloor = \left\lfloor \frac{p \cdot \alpha}{p+1} + \frac{p}{p+1} \right\rfloor = \frac{p \cdot \alpha}{p+1} = \left\lceil \frac{p \cdot \alpha}{p+1} \right\rceil.$$

Otherwise, assume $p \cdot \alpha = k(p+1) + \beta$, where k, β are integer numbers and $1 \leq \beta \leq p$. Hence

$$\left\lceil \frac{p \cdot \alpha}{p+1} \right\rceil = k+1 = \left\lfloor k + \frac{\beta + p}{p+1} \right\rfloor = \left\lfloor \frac{p \cdot \alpha + p}{p+1} \right\rfloor$$

since $1 \leq \frac{\beta+p}{p+1} \leq \frac{2p}{p+1} < 2$.

(iii) By Theorem 2.4(ii), for each independent set A, there exist $p-1$ pairwise disjoint maximum independent sets $S_1, S_2, \ldots, S_{p-1}$ such that $A \cap S_i = \emptyset$ for every $i \in \{1, 2, \ldots, p-1\}$. According to Theorem 1.1, A can be matched into each S_i. Hence, $(p-1) \cdot |A| \leq |N(A)|$.

(iv) By Part (iii) and Theorem 1.6(i), we obtain

$$(k+1) \cdot s_{k+1} \leq (n - p \cdot k) \cdot s_k, 0 \leq k < \alpha.$$

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(v) By Part (iii) and Theorem 1.6(ii), we get that if $n \leq (p+1)\cdot\alpha - p$, then $s_r \geq s_{r+1} \geq \cdots \geq s_\alpha$, where $r = \left\lceil \frac{n-1}{p+1} \right\rceil$. □

Note that, in the case $p = 2$, Theorem 2.6 was proved in [16].

Theorem 2.6 (iii) claims that every graph in the class $\mathbf{W}_p$ is $(p-1)$-quasi-regularizable. In Section 4, we will show that clique corona graphs $G \circ K_p$ are p-quasi-regularizable and belong to the class $\mathbf{W}_p$. It is worth noting, however, that the 5-cycle graph, although a member of $\mathbf{W}_2$, is not 2-quasi-regularizable. These observations, particularly in the context of graphs in the class $\mathbf{W}_p$ with a large number of vertices, naturally lead us to the following.

Conjecture 2.7. *If G is a connected graph in $\mathbf{W}_p$, then G is p-quasi-regularizable whenever $n(G) \geq (p+1)\cdot\alpha(G)$.*

3 The independence polynomial of clique corona graphs

Motivated by a result of Gutman [7] concerning compound graphs, we consider a graph G with the vertex set $V(G) = \{v_1, v_2, \ldots, v_n\}$. For each $i = 1, 2, \ldots, n$, let R_i be pairwise distinct rooted graphs, where the root of R_i is denoted by r_i. The compound graph $G[R_1, R_2, \ldots, R_n]$ is then constructed by identifying each vertex v_i of G with the root r_i of R_i, with this identification carried out simultaneously for all $i = 1, 2, \ldots, n$. For each i, let R_i° denote the graph obtained from R_i by deleting the root vertex r_i and all the edges incident to it. Let $R_i^\bullet$ denote the graph obtained from R_i by deleting the root vertex r_i, all vertices adjacent to r_i, and all edges incident to these vertices. For an independent vertex set $I \subseteq V(G)$, define

$$\omega_i(I) = \begin{cases} I(R_i^\circ; x) & \text{if } v_i \notin I, \\ I(R_i^\bullet; x) & \text{if } v_i \in I. \end{cases}$$

Theorem 3.1 ([7, Theorem 2])**.** *The independence polynomial of $G[R_1, R_2, \ldots, R_n]$ is given by*

$$I(G[R_1, R_2, \ldots, R_n]; x) = \sum_{I \in \mathrm{Ind}(G)} x^{|I|} \prod_{i=1}^{n} \omega_i(I).$$

Building on this result, we describe the independence polynomial of the clique corona graph $G \circ \mathcal{H}$, where $\mathcal{H} = \{K_{p_1}, K_{p_2}, \ldots, K_{p_n}\}$ is a family of complete graphs with $p_i \geq 1$ for each i. For every $i = 1, 2, \ldots, n$, assume that each vertex of K_{p_i} is adjacent to the corresponding vertex v_i in G. Under this construction, the clique corona graph can be viewed as a special case of the compound graph $G[R_1, R_2, \ldots, R_n]$, where each R_i is the complete graph K_{p_i+1} with a designated root vertex. In particular, we have $R_i^\circ = K_{p_i}$ and $R_i^\bullet$ is a trivial graph. Consequently, their independence polynomials are given by

$$I(R_i^\circ; x) = 1 + p_i x \quad \text{and} \quad I(R_i^\bullet; x) = 1.$$

Applying Theorem 3.1, we obtain the following result for clique corona graphs.

Theorem 3.2. *Let G be a graph with vertex set $V(G) = \{v_1, v_2, \ldots, v_n\}$, and let $\mathcal{H} = \{K_{p_1}, K_{p_2}, \ldots, K_{p_n}\}$, where $p_i \geq 1$ for each i. Then*

$$I(G \circ \mathcal{H}; x) = \sum_{I \in \mathrm{Ind}(G)} x^{|I|} \prod_{v_i \notin I} (1 + p_i x).$$

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Proof. For each i, let $R_i = K_{p_i+1}$ and choose one vertex of R_i as the root r_i. Then $G \circ \mathcal{H}$ is the compound graph $G[R_1, R_2, \ldots, R_n]$. Moreover, $R_i^\circ = K_{p_i}$ and $R_i^\bullet$ is a trivial graph. Hence

$$I(R_i^\circ; x) = 1 + p_i x \quad \text{and} \quad I(R_i^\bullet; x) = 1.$$

The conclusion follows immediately from Theorem 3.1. □

For a subset $A \subseteq V(G)$, define $p(A) = \prod_{v_i \in A} p_i$, with the convention $p(\varnothing) = 1$. Expanding the product in Theorem 3.2 and regrouping terms according to the vertices chosen from the attached cliques, we obtain the following.

Theorem 3.3. *The independence polynomial of the clique corona graph $G \circ \mathcal{H}$ is given by:*

$$I(G \circ \mathcal{H}; x) = \sum_{A \subseteq V(G)} p(A) x^{|A|} I(G - A; x). \tag{3.1}$$

Proof. Indeed, by Theorem 3.2,

$$\begin{aligned} I(G \circ \mathcal{H}; x) &= \sum_{I \in \mathrm{Ind}(G)} x^{|I|} \sum_{A \subseteq V(G) \setminus I} p(A) x^{|A|} \\ &= \sum_{A \subseteq V(G)} p(A) x^{|A|} \sum_{\substack{I \in \mathrm{Ind}(G) \\ I \cap A = \varnothing}} x^{|I|} \\ &= \sum_{A \subseteq V(G)} p(A) x^{|A|} I(G - A; x), \end{aligned}$$

as claimed. □

As previously stated, $s_m(G)$ is the coefficient of the degree m term in the independent polynomial of G. Recall that the clique corona graphs $G \circ \mathcal{H}$ are well-covered [25], and $\alpha(G \circ \mathcal{H}) = n$ [5]. Hence, $I(G \circ \mathcal{H}; x) = \sum_{m=0}^{n} s_m(G \circ \mathcal{H}) x^m$.

We note that the independence polynomial of a clique corona graph is expressible in terms of the clique sizes and the independence polynomial of G. One needs to take into account the number of independent sets each vertex of G belongs to. In addition, the bijection of $V(G)$ to $\{p_1, p_2, ..., p_n\}$ is of importance. To be convinced of this, consider $G = P_3$ and observe that the independence polynomial differs if a large clique is attached to the central vertex or to one of the endpoints. For example,

$$\begin{aligned} I(P_3 \circ \{K_1, K_{10}, K_1\}; x) = 1 + 5x + 46x^2 + \mathbf{41}x^3 \neq \\ 1 + 15x + 46x^2 + \mathbf{32}x^3 = I(P_3 \circ \{K_{10}, K_1, K_1\}; x). \end{aligned}$$

Nevertheless, there exist formulas that express the coefficients of the independence polynomial of clique corona graphs in the general case.

Corollary 3.4.

$$s_m(G \circ \mathcal{H}) = \sum_{A \subseteq V(G)} p(A)\, s_{m-|A|}(G - A) = \sum_{j=0}^{m} \sum_{\substack{A \subseteq V(G) \\ |A| = j}} p(A)\, s_{m-j}(G - A) \tag{3.2}$$

$$= \sum_{j=0}^{m} \sum_{1 \le i_1 < i_2 < \cdots < i_j \le n} p(A_{i_1,\ldots,i_j})\, s_{m-j}(G - A_{i_1,\ldots,i_j}) \tag{3.3}$$

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where $s_q(H) = 0$, *whenever* $q < 0$ *or* $q > \alpha(H)$.

Proof. By (3.1),

$$I(G \circ \mathcal{H}; x) = \sum_{A \subseteq V(G)} p(A) x^{|A|} \sum_{q \geq 0} s_q(G - A) x^q = \sum_{A \subseteq V(G)} \sum_{q \geq 0} p(A)\, s_q(G - A) x^{q+|A|}.$$

Comparing the coefficients of x^m on both sides, we obtain

$$s_m(G \circ \mathcal{H}) = \sum_{A \subseteq V(G)} p(A)\, s_{m-|A|}(G - A).$$

Grouping the subsets A according to their cardinality yields the second equality. Finally, if $A = \{v_{i_1}, v_{i_2}, \ldots, v_{i_j}\}$, denoted by $A_{i_1,i_2,\ldots,i_j}$, then the same identity can be rewritten as (3.3). The convention that $s_q(H) = 0$ for $q < 0$ or $q > \alpha(H)$ completes the proof. □

The next example explicitly records the first coefficients of the clique corona independence polynomial.

Example 3.5. Applying (3.3) with $m = 0, 1, 2, 3$, we obtain

$$s_0(G \circ \mathcal{H}) = 1,$$

$$s_1(G \circ \mathcal{H}) = s_1(G) + \sum_{i=1}^{n} p_i,$$

$$s_2(G \circ \mathcal{H}) = s_2(G) + \sum_{i=1}^{n} p_i\, s_1(G - v_i) + \sum_{1 \leq i < j \leq n} p_i p_j,$$

and

$$s_3(G \circ \mathcal{H}) = s_3(G) + \sum_{i=1}^{n} p_i\, s_2(G - v_i) + \sum_{1 \leq i < j \leq n} p_i p_j\, s_1(G - \{v_i, v_j\}) + \sum_{1 \leq i < j < k \leq n} p_i p_j p_k.$$

If all $p_i = p$ for all $1 \leq i \leq n$, we obtain the following.

Corollary 3.6. *If* G *is a graph of order* n, *and* $I(G; x) = \sum_{j=0}^{\alpha(G)} s_j x^j$, *then*

$$I(G \circ K_p; x) = \sum_{m=0}^{n} \left(\sum_{j=0}^{\min\{m, \alpha(G)\}} \binom{n-j}{m-j} p^{m-j} s_j \right) x^m.$$

Proof. By Theorem 3.2, we obtain the following

$$\begin{aligned} I(G \circ K_p; x) &= \sum_{I \in \mathrm{Ind}(G)} x^{|I|} (1 + px)^{n-|I|} \\ &= \sum_{j=0}^{\alpha(G)} s_j x^j (1 + px)^{n-j} \\ &= \sum_{j=0}^{\alpha(G)} \sum_{k=0}^{n-j} \binom{n-j}{k} s_j p^k x^{k+j}. \end{aligned}$$

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Set $m = k + j$. Since $0 \le j \le \alpha(G)$ and $0 \le k = m - j \le n - j$, it follows that $0 \le j \le \min\{m, \alpha(G)\}$. Therefore, the above sum can be rewritten as follows

$$I(G \circ K_p; x) = \sum_{m=0}^{n} \left(\sum_{j=0}^{\min\{m,\alpha(G)\}} \binom{n-j}{m-j} p^{m-j} s_j \right) x^m. \qquad \square$$

In the case $p = 1$, the above corollary was shown in [13, Theorem 2.1]. Furthermore, the unimodality of the independence polynomial of $G \circ K_1$ was proven in [15] for all graphs G satisfying $\alpha(G) \le 4$.

To continue the coefficient analysis in the non-uniform case, for every subset $B \subseteq V(G)$ and every integer $r \ge 0$, let

$$e_r^{(B)} = \sum_{\substack{A \subseteq V(G) - B \\ |A| = r}} p(A),$$

with the convention $e_r^{(B)} = 0$ whenever $r < 0$ or $r > |V(G) - B|$. In particular, $e_r = e_r^{(\varnothing)}$ is the r-th elementary symmetric sum in $p_1, p_2, \ldots, p_n$. Recall that $\Omega_j(G)$ denotes the family of all independent sets of cardinality j in G.

Proposition 3.7. *For every integers $r \ge 0$ and $j \ge 0$,*

$$\sum_{S \in \Omega_j(G)} e_r^{(S)} = \sum_{F \subseteq E(G)} (-1)^{|F|} \binom{n - r - |V(F)|}{j - |V(F)|} e_r^{(V(F))}, \tag{3.4}$$

where $V(F)$ is the set of end-vertices of the edges in F, and $\binom{a}{b} = 0$ whenever $b < 0$ or $b > a$. Consequently,

$$s_m(G \circ \mathcal{H}) = \sum_{j=0}^{\min\{m,\alpha(G)\}} \sum_{F \subseteq E(G)} (-1)^{|F|} \binom{n - m + j - |V(F)|}{j - |V(F)|} e_{m-j}^{(V(F))}. \tag{3.5}$$

Proof. By (3.3),

$$s_m(G \circ \mathcal{H}) = \sum_{j=0}^{\min\{m,\alpha(G)\}} \sum_{S \in \Omega_j(G)} e_{m-j}^{(S)}.$$

Hence it is enough to prove (3.4). Fix r and j. Since

$$\sum_{S \in \Omega_j(G)} e_r^{(S)} = \sum_{\substack{A \subseteq V(G) \\ |A| = r}} p(A)\, s_j(G - A),$$

we count the j-subsets of $V(G) - A$ containing no edge of $G - A$ by inclusion-exclusion over $E(G - A)$. This gives

$$s_j(G - A) = \sum_{F \subseteq E(G-A)} (-1)^{|F|} \binom{n - r - |V(F)|}{j - |V(F)|}.$$

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Multiplying by $p(A)$, summing over all $A \subseteq V(G)$ with $|A| = r$, and then changing the order of summation, we obtain

$$\sum_{S\in\Omega_j(G)} e_r^{(S)} = \sum_{F\subseteq E(G)} (-1)^{|F|} \binom{n-r-|V(F)|}{j-|V(F)|} \sum_{\substack{A\subseteq V(G)-V(F)\\ |A|=r}} p(A)$$
$$= \sum_{F\subseteq E(G)} (-1)^{|F|} \binom{n-r-|V(F)|}{j-|V(F)|} e_r^{(V(F))},$$

which proves (3.4). Substituting $r = m - j$ yields (3.5). □

Example 3.8. Let $\mathcal{P}_3(G)$ and $\mathcal{T}_3(G)$ denote, respectively, the families of subgraphs of G isomorphic to P_3 and K_3. Taking $j = 2$ and $j = 3$ in (3.4), we obtain

$$\sum_{S\in\Omega_2(G)} e_r^{(S)} = \binom{n-r}{2} e_r - \sum_{uv\in E(G)} e_r^{(u,v)},$$

and

$$\sum_{S\in\Omega_3(G)} e_r^{(S)} = \binom{n-r}{3} e_r - (n-r-2) \sum_{uv\in E(G)} e_r^{(u,v)} + \sum_{P\in\mathcal{P}_3(G)} e_r^{(V(P))} - \sum_{T\in\mathcal{T}_3(G)} e_r^{(V(T))}.$$

4 Partial unimodality of the independence polynomial of $G \circ K_p$

In this section, we focus on the unimodality of the independence polynomial of the clique corona graph $G \circ K_p$, which also generalizes some related results.

Lemma 4.1. *Given two positive real numbers x and y, if $x \geq y$, then either $\lfloor x \rfloor \geq \lceil y \rceil$ or $\lfloor x \rfloor < \lceil y \rceil$ and $\lceil y \rceil - \lfloor x \rfloor = 1$.*

Proof. Since $x \geq y$, we know that $\lfloor x \rfloor \geq \lfloor y \rfloor$. Assume that $\lfloor x \rfloor < \lceil y \rceil$. Then $\lfloor y \rfloor \leq \lfloor x \rfloor < \lceil y \rceil$. Clearly,

$$\lceil y \rceil - \lfloor y \rfloor = \begin{cases} 0, & \text{if } y \text{ is integer} \\ 1, & \text{otherwise.} \end{cases}$$

Hence, y is not integer. Thus $\lceil y \rceil - \lfloor y \rfloor = 1$, and, consequently, $\lceil y \rceil - 1 \leq \lfloor x \rfloor < \lceil y \rceil$, which means $\lceil y \rceil - 1 = \lfloor x \rfloor$. □

The corresponding result for connected graphs and $p = 2$ was proved in [16].

Theorem 4.2. *Let G be a graph of order n, not necessarily connected, and let $p \geq 2$. If $I(G \circ K_p; x) = \sum_{k=0}^{n} t_k x^k$ then the following assertions hold:*

(i) *$G \circ K_p$ is in $\mathbf{W}_p$ with $\alpha(G \circ K_p) = n$;*

(ii) *$G \circ K_p$ is p-quasi-regularizable with $n(G \circ K_p) = (p+1) \cdot n$;*

(iii) *$p \cdot (n-k) \cdot t_k \leq (k+1) \cdot t_{k+1} \leq (p+1) \cdot (n-k) \cdot t_k$, for all $0 \leq k < n$;*

(iv) *$t_0 \leq t_1 \leq \cdots \leq t_{\lfloor (n+1)\frac{p}{p+1} \rfloor}$ and $t_{\lceil \frac{(p+1)\cdot n-1}{p+2} \rceil} \geq \cdots \geq t_{n-1} \geq t_n$;*

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(v) *If* $(p+1)(p+2) \geq n+1$, *then the sequence* $t_0, t_1, \cdots, t_n$ *is unimodal;*

(vi) *In particular, if* $p \geq n$, *then* $t_0 \leq t_1 \leq \cdots \leq t_n$.

Proof. (i) It follows from Corollary 2.3.

(ii) Let $H = G \circ K_p$, and let $S = S_1 \cup S_2$ be an independent set in H, where $S_1 \subseteq V(G)$, while $S_2 \subseteq V(H) - V(G)$. Then

$$p \cdot |S_1| = |N_H(S_1) - V(G)| \leq |N_H(S_1)|,$$

because every vertex of S_1 has exactly p neighbors belonging to $V(H) - V(G)$, while $p \cdot |S_2| = |N_H(S_2)|$, since each vertex from S_2 has exactly p neighbors in H, while $N_H(v) \cap N_H(w) = \emptyset$ for every two distinct vertices from S_2. Hence, we get that:

$$p \cdot |S| = p \cdot |S_1| + p \cdot |S_2| \leq |N_H(S_1) - V(G)| + |N_H(S_2)| \leq |N_H(S)|.$$

This concludes that H is p-quasi-regularizable.

By definition of clique corona graph $G \circ K_p$, the vertex set $V(H)$ can be partitioned into n copies of K_{p+1}, where each K_{p+1} shares exactly one vertex with G and its remaining p vertices form a copy of K_p. Therefore,

$$n(H) = (p+1) \cdot n.$$

(iii) For $0 \leq k < n$, let $\mathcal{I}_k(H)$ denote the family of all independent sets of cardinality k in H, and set

$$\mathcal{P}_k = \{(S, U) : S \in \mathcal{I}_k(H),\ U \in \mathcal{I}_{k+1}(H),\ S \subset U\}.$$

As noted in Part (ii), the vertex set of H is partitioned into n cliques of order $p+1$, one for each vertex of G. Every independent set $S \in \mathcal{I}_k(H)$ meets exactly k of these cliques. In each of the remaining $n-k$ cliques, all p vertices belonging to the attached copy of K_p can be added to S, while at most $p+1$ vertices of that clique can be added to S. Hence, the number of sets $U \in \mathcal{I}_{k+1}(H)$ containing S lies between $p(n-k)$ and $(p+1)(n-k)$. On the other hand, every $U \in \mathcal{I}_{k+1}(H)$ contains exactly $k+1$ independent subsets of cardinality k. Double-counting the pairs in $\mathcal{P}_k$ gives

$$p(n-k)t_k \leq |\mathcal{P}_k| = (k+1)t_{k+1} \leq (p+1)(n-k)t_k, \qquad 0 \leq k < n,$$

which proves Part (iii).

(iv) The left-hand inequality in Part (iii) implies $t_k \leq t_{k+1}$ whenever $p(n-k) \geq k+1$, equivalently, $k \leq (pn-1)/(p+1)$. Since

$$\left\lfloor \frac{p(n+1)}{p+1} \right\rfloor - 1 = \left\lfloor \frac{pn-1}{p+1} \right\rfloor,$$

we obtain

$$t_0 \leq t_1 \leq \cdots \leq t_{\lfloor \frac{p(n+1)}{p+1} \rfloor}.$$

The right-hand inequality in Part (iii) implies $t_{k+1} \leq t_k$ whenever $(p+1)(n-k) \leq k+1$, equivalently,

$$k \geq \left\lceil \frac{n(p+1)-1}{p+2} \right\rceil.$$

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Consequently,

$$t_r \geq t_{r+1} \geq \cdots \geq t_n, \qquad r = \left\lceil \frac{n(p+1)-1}{p+2} \right\rceil \leq n.$$

(v) It is easy to see that

$$(n+1)\frac{p}{p+1} \geq (n+1)\frac{p+1}{p+2} - 1$$

if and only if

$$(p+1)(p+2) \geq n+1.$$

Assume that $(p+1)(p+2) \geq n+1$.
By Part (iv), if

$$\left\lfloor (n+1)\frac{p}{p+1} \right\rfloor \geq \left\lceil \frac{(p+1)\cdot n - 1}{p+2} \right\rceil = \left\lceil (n+1)\frac{p+1}{p+2} \right\rceil - 1,$$

then the sequence $t_0, t_1, \cdots, t_n$ is unimodal and all the modes are located between $\left\lceil \frac{(p+1)\cdot n-1}{p+2} \right\rceil$ and $\left\lfloor (n+1)\frac{p}{p+1} \right\rfloor$.

On the other hand, if $\left\lfloor (n+1)\frac{p}{p+1} \right\rfloor \leq \left\lceil \frac{(p+1)\cdot n-1}{p+2} \right\rceil$, then, by Lemma 4.1,

$$\left\lfloor (n+1)\frac{p}{p+1} \right\rfloor + 1 = \left\lceil \frac{(p+1)\cdot n - 1}{p+2} \right\rceil.$$

Thus either one of $\left\lfloor (n+1)\frac{p}{p+1} \right\rfloor$ and $\left\lceil \frac{(p+1)\cdot n-1}{p+2} \right\rceil$ is a mode or both of these numbers are modes.

(vi) Clearly, $n+1 > (n+1)\frac{p}{p+1}$. In addition, it should be noted that if $p \geq n$, then $(n+1)\frac{p}{p+1} \geq n$. Hence, by Part (iv), we have

$$t_0 \leq t_1 \leq \cdots \leq t_n,$$

as claimed. □

Example 4.3. (i) With reference to Theorem 4.2(iv) concerning $I(C_5 \circ K_1; x)$, we obtain $\left\lfloor \frac{(\alpha+1)p}{p+1} \right\rfloor = \left\lfloor \frac{(5+1)\cdot 1}{1+1} \right\rfloor = 3$ and $\left\lceil \frac{(p+1)\alpha-1}{p+2} \right\rceil = \left\lceil \frac{(1+1)5-1}{1+2} \right\rceil = 3$. Thus, $I(C_5 \circ K_1; x)$ is unimodal with the mode occurring at 3.

It can also be seen from the explicit form of the polynomial

$$I(C_5 \circ K_1; x) = 1 + 10x + 35x^2 + \mathbf{55}x^3 + 40x^4 + 11x^5.$$

(ii) Based on Theorem 4.2(iv) for $I(C_7 \circ K_1; x)$, we have $\left\lfloor \frac{(\alpha+1)p}{p+1} \right\rfloor = \left\lfloor \frac{(7+1)\cdot 1}{1+1} \right\rfloor = 4$ and $\left\lceil \frac{(p+1)\alpha-1}{p+2} \right\rceil = \left\lceil \frac{(1+1)7-1}{1+2} \right\rceil = 5$. Thus, $I(C_7 \circ K_1; x)$ is unimodal with the mode occurring at either 4 or 5. It can also be seen from the explicit form of the polynomial

$$I(C_7 \circ K_1; x) = 1 + 14x + 77x^2 + 217x^3 + \mathbf{343}x^4 + 308x^5 + 147x^6 + 29x^7,$$

which is unimodal with mode 4.

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The following proposition gives us the unimodality of $I(G \circ K_p; x)$ with mode either $\left\lfloor \frac{(n(G)+1)\cdot p}{p+1} \right\rfloor$ or $\left\lceil \frac{(n(G)+1)\cdot p}{p+1} \right\rceil$, whenever G is quasi-regularizable and $\alpha(G) \leq p+1$.

Proposition 4.4. *If G is a quasi-regularizable graph and $\alpha(G) \leq p+1$, then the inequalities $t_m \geq t_{m+1} \geq \cdots \geq t_{n(G)}$ hold for all $m \geq \left\lceil \frac{(n(G)+1)\cdot p}{p+1} \right\rceil$, where $I(G \circ K_p; x) = \sum_{k=0}^{n(G)} t_k \cdot x^k$. In particular, $I(G \circ K_p; x)$ is unimodal with*

$$\left\lfloor \frac{(n(G)+1)\cdot p}{p+1} \right\rfloor \leq \operatorname{mode}(I(G \circ K_p; x)) \leq \left\lceil \frac{(n(G)+1)\cdot p}{p+1} \right\rceil.$$

Proof. Let S be a maximum independent set of G. Since G is quasi-regularizable, we see that

$$2 \cdot |S| \leq |S| + |N(S)| \leq |V(G)| = n(G).$$

Hence, $\alpha(G) \leq \lfloor \frac{n(G)}{2} \rfloor$. Moreover, we always have $\frac{(n(G)+1)\cdot p}{p+1} > \frac{n(G)}{2}$. Thus,

$$\left\lfloor \frac{(n(G)+1)\cdot p}{p+1} \right\rfloor \geq \lfloor \frac{n(G)}{2} \rfloor \geq \alpha(G).$$

Hence, by Corollary 3.6 for all $m \geq \left\lceil \frac{(n(G)+1)\cdot p}{p+1} \right\rceil$, we have

$$t_m = \sum_{j=0}^{m} \binom{n(G)-j}{m-j} \cdot p^{m-j} \cdot s_j = \sum_{j=0}^{\alpha(G)} \binom{n(G)-j}{m-j} \cdot p^{m-j} \cdot s_j.$$

So, we infer that

$$\begin{aligned} t_m - t_{m+1} &= \sum_{j=0}^{\alpha(G)} \left[\binom{n(G)-j}{m-j} - \binom{n(G)-j}{m+1-j} p \right] \cdot p^{m-j} \cdot s_j \\ &= \sum_{j=0}^{\alpha(G)} [(1+p)\cdot m + 1 - j - p \cdot n(G)] \frac{(n(G)-j)! \cdot p^{m-j} \cdot s_j}{(m+1-j)! \cdot (n(G)-m)!}. \end{aligned}$$

By the assumption, we have $m \geq \frac{(n(G)+1)\cdot p}{p+1}$. This implies that $(p+1)\cdot m \geq n(G)\, p + p$. Hence, $(p+1)\cdot m - n(G)\cdot p - p \geq 0$.

If $\alpha(G) \leq p+1$, then $j \leq \alpha(G) \leq p+1$ which implies that $1 - j \geq -p$. Therefore,

$$(1+p)\cdot m + 1 - j - p \cdot n(G) \geq (p+1)\cdot m - n(G)\cdot p - p \geq 0.$$

We conclude that $t_m \geq t_{m+1}$. On the other word, we obtain the following inequalities

$$t_{\left\lceil \frac{(n(G)+1)\cdot p}{p+1} \right\rceil} \geq \cdots \geq t_{n(G)-1} \geq t_{n(G)}.$$

Together with Theorem 4.2(iv), we have

$$t_0 \leq t_1 \leq \cdots \leq t_{\lfloor (n(G)+1)\frac{p}{p+1} \rfloor}.$$

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Thus, $I(G \circ K_p; x)$ is unimodal with

$$\left\lfloor \frac{(n(G)+1)\cdot p}{p+1} \right\rfloor \leq \text{mode}(I(G \circ K_p; x)) \leq \left\lceil \frac{(n(G)+1)\cdot p}{p+1} \right\rceil,$$

as claimed. □

Example 4.5. (i) $I(C_5 \circ K_2; x) = 1 + 15x + 85x^2 + 230x^3 + \mathbf{300}x^4 + 152x^5$ is unimodal with mode 4. Comparing with Theorem 4.2 *(iv)*, we have

$$\left\lfloor \frac{(\alpha+1)p}{p+1} \right\rfloor = \left\lfloor \frac{(5+1)2}{2+1} \right\rfloor = 4 = \left\lceil \frac{(\alpha+1)p}{p+1} \right\rceil$$

and $\left\lceil \frac{(p+1)\alpha-1}{p+2} \right\rceil = \left\lceil \frac{(2+1)5-1}{2+2} \right\rceil = 4.$

(ii) $I(C_7 \circ K_2; x) = 1+21x+182x^2+847x^3+2296x^4+\mathbf{3640}x^5+3136x^6+1136x^7$ is unimodal with mode 5.

Comparing with Theorem 4.2(iv), we have $\left\lfloor \frac{(\alpha+1)p}{p+1} \right\rfloor = \left\lfloor \frac{(7+1)2}{2+1} \right\rfloor = 5,$

and $\left\lceil \frac{(\alpha+1)p}{p+1} \right\rceil = \left\lceil \frac{(7+1)2}{2+1} \right\rceil = 6$, while $\left\lceil \frac{(p+1)\alpha-1}{p+2} \right\rceil = \left\lceil \frac{(2+1)7-1}{2+2} \right\rceil = 5.$

Now, we are in a position to conclude with the main result of this section.

Theorem 4.6. (i) *If* $(p+1)(p+2) \geq n(G)+1$, *then* $I(G \circ K_p; x)$ *is unimodal with*

$$\left\lceil \frac{(p+1)\cdot n(G)-1}{p+2} \right\rceil \leq \text{mode}(I(G \circ K_p; x)) \leq \left\lfloor \frac{(n(G)+1)\cdot p}{p+1} \right\rfloor.$$

(ii) *If* $(p+1)(p+2) < n(G)+1$, *G is a quasi-regularizable graph, and* $\alpha(G) \leq p+1$, *then* $I(G \circ K_p; x)$ *is unimodal with*

$$\left\lfloor \frac{(n(G)+1)\cdot p}{p+1} \right\rfloor \leq \text{mode}(I(G \circ K_p; x)) \leq \left\lceil \frac{(n(G)+1)\cdot p}{p+1} \right\rceil.$$

(iii) *If* $p \geq n(G)$, *then* $I(G \circ K_p; x)$ *is unimodal with mode* $n(G)$.

Proof. (i) It follows from Theorem 4.2(iv), (v).

(ii) It follows from Theorem 4.2(iv) and Proposition 4.4,
because $n(G)+1 > (p+1)(p+2)$ implies $\left\lceil \frac{n(G)\cdot(p+1)-1}{p+2} \right\rceil > \left\lceil \frac{(n(G)+1)\cdot p}{p+1} \right\rceil$. ž

(iii) It is exactly the claim of Theorem 4.2(vi). □

The conjecture about the unimodality of $I(G \circ K_1; x)$ for all graphs G was stated in [10, Conjecture 3.3], validated it in the case $\alpha(G) \leq 4$ in [10] and extended to $5 \leq \alpha(G) \leq 8$ in [**?**].

Corollary 4.7. *Suppose* G *is quasi-regularizable. If* $\alpha(G) \leq 4$, *then* $I(G \circ K_p; x)$ *is unimodal for all* $p \geq 1$.

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Proof. If $\alpha(G) \leq 3$, by Theorem 4.6(i),(ii), $I(G \circ K_p; x)$ is unimodal for all $p \geq 2$. Together with [10, Theorems 2.1 and 2.2], we obtain $I(G \circ K_p; x)$ is unimodal for all $p \geq 1$.

If $\alpha(G) = 4$, by Theorem 4.6(i),(ii) again, then $I(G \circ K_p; x)$ is unimodal for all $p \geq 3$. Now we claim that $I(G \circ K_2; x)$ is also unimodal. Indeed, assume that $I(G \circ K_2; x) = \sum_{m=0}^{n(G)} t_m x^m$ and $I(G; x) = s_0 + s_1 x + \cdots + s_4 x^4$. Since $t_m = \sum_{j=0}^{m} \binom{n(G)-j}{m-j} 2^{m-j} s_j$, we know that $t_0 \leq t_1 \leq \cdots \leq t_m$ for all $m \leq \left\lfloor \frac{2(n(G)+1)}{3} \right\rfloor$, by Theorem 4.2(iv). Now, we consider the two following cases:

Case 1: $n(G) = 3s-1$ or $n(G) = 3s$, where s is a positive integer. Hence $\left\lfloor \frac{2(n(G)+1)}{3} \right\rfloor = 2s$. Therefore, $t_0 \leq t_1 \leq \cdots \leq t_{2s}$. However, in this case, if $m \geq 2s+1$, then $3(m+1-2s) \geq 6$. For $0 \leq j \leq \alpha(G) = 4$, we have $3(m+1-2s) - j \geq 2$. By the proof of Proposition 4.4, $t_m \geq t_{m+1}$ for all $m \geq 2s+1$ which also means $t_{2s+1} \geq t_{2s+2} \geq \cdots \geq t_{n(G)}$. Hence, $I(G \circ K_2; x)$ is unimodal.

Case 2: $n(G) = 3s+1$ ($s \in \mathbb{N}$). Hence $\left\lfloor \frac{2(n(G)+1)}{3} \right\rfloor = 2s+1$. In this case, we have $t_0 \leq t_1 \leq \cdots \leq t_{2s+1}$. If $m \geq 2s+2$, then $3(m+1-2s) \geq 9$. For $0 \leq j \leq \alpha(G) = 4$, we have $3(m+1-2s) - j \geq 5$. Thus, $t_m \geq t_{m+1}$ for all $m \geq 2s+1$ which means $t_{2s+2} \geq t_{2s+3} \geq \cdots \geq t_{n(G)}$. Hence, $I(G \circ K_2; x)$ is unimodal. □

5 Application to the corona of a star and a complete graph

In this section, we settle the unimodality of the independence polynomial of the corona of a star graph and a complete graph on p vertices, with $p \geq 1$.

Lemma 5.1. *If $P(x)$ is unimodal with the mode m, then the polynomial $(1+px) \cdot P(x)$ is unimodal with mode m or $m+1$.*

Proof. Let $P(x) = \sum_{k=0}^{n} s_k x^k$. We have

$$\begin{aligned}(1+px) \cdot P(x) &= \sum_{k=0}^{n} s_k \cdot x^k + \sum_{k=0}^{n} p \cdot s_k \cdot x^{k+1} \\ &= s_0 + \sum_{k=1}^{n} [s_k + p \cdot s_{k-1}] x^k + p \cdot s_n \cdot x^{n+1}.\end{aligned}$$

Let $t_k = s_k + p s_{k-1}$. Since $P(x)$ is unimodal with the mode m, we have that $s_0 \leq \cdots \leq s_m \geq \cdots \geq s_n$. It is clear that

$$s_0 \leq t_1 \leq \cdots \leq t_m \quad \text{and} \quad t_{m+1} \geq \cdots \geq t_n \geq p s_n,$$

as required. □

Lemma 5.2. *Let p and n be positive integers. If $p \geq \sqrt{n+1} - 2$, then*

$$\left\lceil \frac{np}{p+1} \right\rceil + 1 \geq \frac{n(p+1)-1}{p+2}.$$

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Proof. Assume that $p+1$ is a divisor of n, then $n = k(p+1)$ for some integer k. Then we have $\left\lceil \frac{np}{p+1} \right\rceil = kp$. By the assumption, $p \geq \sqrt{n+1}-2$ which implies that $n \leq p^2+4p+3$. Hence, $p+3 \geq k$. This implies that

$$(kp+1)(p+2) \geq k(p+1)^2 - 1,$$

equivalently, we obtain $kp+1 \geq \frac{n(p+1)-1}{p+2}$. Therefore, $\left\lceil \frac{np}{p+1} \right\rceil + 1 \geq \frac{n(p+1)-1}{p+2}$.

Now we assume $n = k(p+1)+s$ where $1 \leq s \leq p$. By the assumption, $p \geq \sqrt{n+1}-2$ which implies that $n \leq p^2+4p+3$. Hence, $k(p+1)+s \leq p^2+4p+3$. We claim that $p+s+3 \geq k$. Assume, to the contrary, that $p+s+3 < k$. We have

$$\begin{aligned}(p+s+3)(p+1)+s &< k(p+1)+s \\ &\leq p^2+4p+3.\end{aligned}$$

This is equivalent to $sp+2s < 0$, a contradiction.

Since $p+s+3 \geq k$, so

$$(kp+s+1)(p+2) \geq (kp+k+s)(p+1) - 1.$$

Therefore,

$$kp+s+1 \geq \frac{(kp+k+s)(p+1)-1}{p+2}.$$

We note that

$$\left\lceil \frac{np}{p+1} \right\rceil = \left\lceil kp+s-\frac{s}{p+1} \right\rceil = kp+s.$$

So $\left\lceil \frac{np}{p+1} \right\rceil + 1 \geq \frac{n(p+1)-1}{p+2}$, and this completes the proof. □

Remark 5.3. By the assumption, $kp+s+1 < \frac{n(p+1)-1}{p+2}$. Thus, $s+p+3 < k$. On the other hand, since $n \leq p^2+4p+3$, we get $k(p+1)+s \leq p^2+4p+3$. Together with the condition above, we have $(s+p+3)(p+1)+s < p^2+4p+3$. This is equivalent to $sp+2s < 0$, a contradiction. Therefore, there is no integer in the interval $\left[\left\lceil \frac{np}{p+1} \right\rceil + 1, \frac{n(p+1)-1}{p+2}\right)$.

A *star* S_n is the complete bipartite graph $K_{1,n}$. Suppose that bipartition of S_n is $(\{v_1,\ldots,v_n\};\{v_{n+1}\})$. As a main result in this section, we show that the independence polynomial of the corona of S_n and K_p is unimodal for all $p \geq \sqrt{n+1}-2$.

Theorem 5.4. *$I(S_n \circ K_p; x)$ is unimodal for all $p \geq \sqrt{n+1}-2$. Moreover,*

(i) *if $p \geq n-1$, then*

$$\left\lceil \frac{(p+1)\cdot(n+1)-1}{p+2} \right\rceil \leq \operatorname{mode}(I(S_n \circ K_p; x)) \leq \left\lfloor (n+2)\,\frac{p}{p+1} \right\rfloor.$$

(ii) *if $\sqrt{n+1}-2 \leq p \leq n-2$, then*

$$\left\lfloor \frac{np}{p+1} \right\rfloor + 1 \leq \operatorname{mode}(I(S_n \circ K_p; x)) \leq \left\lceil \frac{np}{p+1} \right\rceil + 2.$$

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Proof. The maximal independent sets of S_n are $\{v_{n+1}\}$ and $\{v_1, v_2, \ldots, v_n\}$. Hence $\alpha(S_n) = n$. If $p \geq n-1$, then $(p+1)(p+2) \geq n+1$. Applying Theorem 4.6(i), $I(S_n \circ K_p; x)$ is unimodal and its mode is located between $\left\lceil \frac{(p+1)\cdot(n+1)-1}{p+2} \right\rceil$ and $\left\lfloor (n+2)\,\frac{p}{p+1} \right\rfloor$.

We now consider its unimodality in case $\sqrt{n+1} - 2 \leq p \leq n-2$. The independence polynomial of S_n is

$$I(S_n; x) = 1 + (n+1)x + \binom{n}{2}x^2 + \binom{n}{3}x^3 + \cdots + \binom{n}{n}x^n = (1+x)^n + x.$$

Denote $L = S_n \circ K_p$. We have $L - v_{n+1} = nK_{p+1} \cup K_p$ and $L - N[v_{n+1}] = nK_p$. By Lemma 1.4,

$$\begin{aligned} I(L; x) &= I(L - v_{n+1}; x) + xI(L - N[v_{n+1}]; x) \\ &= (1+px)(1+(p+1)x)^n + x(1+px)^n \\ &= (1+px)\left[(1+(p+1)x)^n + x(1+px)^{n-1}\right]. \end{aligned}$$

Since, $Q(x) = 1 + px$, from Lemma 5.1, we need to prove $P(x) = (1+(p+1)x)^n + x(1+px)^{n-1}$ unimodal with

$$\left\lfloor \frac{np}{p+1} \right\rfloor + 1 \leq \operatorname{mode}(P(x)) \leq \left\lceil \frac{np}{p+1} \right\rceil + 1$$

to obtain the unimodality of $I(L; x)$. We have

$$\begin{aligned} P(x) &= \sum_{i=0}^{n} \binom{n}{i} (p+1)^i x^i + \sum_{i=0}^{n-1} \binom{n-1}{i} p^i x^{i+1} \\ &= 1 + \sum_{i=1}^{n} \left[\binom{n}{i}(p+1)^i + \binom{n-1}{i-1}p^{i-1} \right] x^i. \end{aligned}$$

Denote $t_i = \binom{n}{i}\left[(p+1)^i + \frac{i}{n}p^{i-1}\right]$. Then

$$\begin{aligned} t_{i+1} &= \binom{n}{i+1}\left[(p+1)^{i+1} + \frac{i+1}{n}p^i\right] \\ &= \binom{n}{i}\left[\frac{n-i}{i+1}(p+1)^{i+1} + \frac{n-i}{n}p^i\right]. \end{aligned}$$

We consider

$$\begin{aligned} \frac{t_{i+1} - t_i}{\binom{n}{i}} &= \left[\frac{n-i}{i+1}(p+1)^{i+1} - (p+1)^i\right] + \left[\frac{n-i}{n}p^i - \frac{i}{n}p^{i-1}\right] \\ &= \left[\frac{(n-i)(p+1)}{i+1} - 1\right](p+1)^i + \frac{p^{i-1}}{n}[(n-i)p - i]. \end{aligned}$$

If $i \leq \frac{np}{p+1}$, then $np - i(p+1) \geq 0$ and thus $(n-i)p - i \geq 0$. Moreover, since $p < n-1$, $\frac{np}{p+1} < \frac{np+n-1}{p+2}$. Hence $i < \frac{np+n-1}{p+2}$. This yields $np + n - 1 > i(p+2)$. It implies that $\frac{(n-i)(p+1)}{i+1} - 1 > 0$. From this, we obtain that $t_{i+1} \geq t_i$ for all $i \leq \left\lfloor \frac{np}{p+1} \right\rfloor$. In other words,

$$t_1 \leq t_2 \leq \cdots \leq t_{m+1},$$

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for all $m \leq \left\lfloor \frac{np}{p+1} \right\rfloor$.

If $i \geq \left\lceil \frac{np}{p+1} \right\rceil + 1$, then $i > \frac{np}{p+1}$. This implies that $i - (n-i)p > 0$. We have

$$\begin{aligned}\frac{t_i - t_{i+1}}{\binom{n}{i}} &= \left[(p+1)^i + \frac{i}{n}p^{i-1}\right] - \left[\frac{n-i}{i+1}(p+1)^{i+1} + \frac{n-i}{n}p^i\right] \\ &= \frac{1}{n(i+1)}\left[n(i+1)(p+1)^i + i(i+1)p^{i-1} - n(n-i)(p+1)^{i+1}\right. \\ &\qquad \left. -(n-i)(i+1)p^i\right].\end{aligned}$$

Therefore, $P(x)$ is unimodal and its mode satisfies

$$\left\lfloor \frac{np}{p+1} \right\rfloor + 1 \leq \text{mode}(P(x)) \leq \left\lceil \frac{np}{p+1} \right\rceil + 1.$$

By the assumption, $i > \frac{np}{p+1}$ and so $i(p+1) > np$. This implies that $i(p+1) - np \geq 1$ and thus $i - (n-i)p \geq 1$. Then

$$i(i+1)p^{i-1} - (n-i)(i+1)p^i = (i+1)p^{i-1}[i - (n-i)p] \geq (i+1)p^{i-1}.$$

On the other hand, by Lemma 5.2, we have $i \geq \frac{n(p+1)-1}{p+2}$. Hence,

$$(p+2)i - np - n + 1 \geq n(p+1) - 1 - np - n + 1 \geq 0.$$

Thus

$$\begin{aligned}n(i+1)(p+1)^i - n(n-i)(p+1)^{i+1} &= n(p+1)^i[i+1-(n-i)(p+1)] \\ &= n(p+1)^i[(p+2)i + 1 - np - n] \geq 0.\end{aligned}$$

Therefore,

$$t_{m+1} \geq t_{m+2} \geq \cdots \geq t_n,$$

for all $m \geq \left\lceil \frac{np}{p+1} \right\rceil$, which completes the proof. □

If $p < \sqrt{n+1} - 2$, then the mode of the polynomial $I(S_n \circ K_p; x)$, as a function of n and p, is quite disordered. It was shown in [11, Theorem 3.1] that $I(S_n \circ K_1; x)$ is unimodal and its mode is unique and equals $2\left\lceil \frac{n}{3} \right\rceil - 1 + [(n-1) \mod 3]$. Furthermore, in the case $n = 24$ and $p = 2$, $\frac{np}{p+1} = 16$ and $\left\lfloor \frac{(n+2)p}{p+1} \right\rfloor = 17$, $\left\lceil \frac{(n+2)p}{p+1} \right\rceil = 18$, whereas $I(S_{24} \circ K_2; x)$ is unimodal with unique mode 19.

6 Conclusions

Briefly, in this paper, we have presented several significant results related to the independence polynomial of the clique corona graph, which can be considered as a typical case of $\mathbf{W}_p$ and p-quasi-regularizable graphs. In Section 3, we also obtained the general coefficient identity

$$s_m(G \circ \mathcal{H}) = \sum_{A \subseteq V(G)} p(A)\, s_{m-|A|}(G - A),$$

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which clarifies the role of the attachment map in the non-uniform clique corona graph. In addition, we have investigated the unimodality of the independence polynomial of clique corona graph $G \circ K_p$ for various classes of graphs, including: G being a quasi-regularizable graph with $\alpha(G) \leq 4$ (Corollary 4.7); G being an arbitrary graph and p satisfying $(p+1)(p+2) \geq n(G)+1$ (Theorem 4.2 *(v)*); and G being a quasi-regularizable graph with $\alpha(G) \leq p+1$ (Proposition 4.4). As an application, we have proved that $I(S_n \circ K_p; x)$ is unimodal for all $p \geq \sqrt{n+1} - 2$. As a conclusion, these findings have led us to the following.

Conjecture 6.1. *If G is an arbitrary graph, then $I(G \circ K_p; x)$ is unimodal for all $p \geq 1$.*

Note that this conjecture can also be confirmed for the class of claw-free graphs by a corresponding result in [15, Corollary 3.12]. Consequently, the conjecture is expected to hold even for graphs that contain a claw.

Another finding concerning the decreasing property of the coefficient sequence of the independence polynomial for perfect graphs was presented in [12]. Recall that a graph G is *perfect* if $\chi(H) = \omega(H)$ for every induced subgraph H of G, where $\chi(H)$ denotes the chromatic number of H [2], and $\omega(H)$ denotes the clique number of H.

Proposition 6.2 ([12])**.** *Let G be a perfect graph, with the independence number $\alpha(G) = \alpha$ and the clique number $\omega(G) = \omega$. Then*

$$s_{\lceil(\omega\alpha-1)/(\omega+1)\rceil} \geq \cdots \geq s_{\alpha-1} \geq s_{\alpha}.$$

By an observation in [14], we know that $G \circ H$ is a perfect graph if and only if both G and H are perfect. Since K_p is perfect for every $p \geq 1$, we can immediately deduce the following result.

Corollary 6.3. *If G is a perfect graph and $I(G \circ K_p; x) = \sum_{m=0}^{n(G)} t_m x^m$, then*

$$t_{\lceil(\omega\alpha-1)/(\omega+1)\rceil} \geq \cdots \geq t_{\alpha-1} \geq t_{\alpha},$$

where $\alpha = \alpha(G \circ K_p) = n(G)$ and $\omega = \omega(G \circ K_p) = \max\{\omega(G), p+1\}$.

This motivates the following.

Problem 6.4. Characterize perfect graphs G such that $I(G \circ K_p; x)$ is unimodal.

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References

[1] Y. Alavi, P. J. Malde, A. J. Schwenk and P. Erdős, The vertex independence sequence of a graph is not constrained, *Congr. Numer.* **58** (1987), 15–23.

[2] C. Berge, Färbung von graphen deren sämtliche bzw. deren ungerade kreise starr sind (zusammenfassung), *Wiss. Z. Martin-Luther-Univ. Halle* **10** (1961), 114–115.

Accepted manuscript

[3] C. Berge, Some common properties for regularizable graphs, edge-critical graphs and b-graphs, *Ann. Discrete Math.* **12** (1982), 31–44.

[4] M. Chudnovsky and P. Seymour, The roots of the independence polynomial of a claw-free graph, *J. Comb. Theory, Ser. B* **97** (2007), 350–357, doi:10.1016/j.jctb.2006.06.001, https://doi.org/10.1016/j.jctb.2006.06.001.

[5] T. H. Do and M. H. Pham, The size of Betti tables of edge ideals of clique corona graphs, *Arch. Math.* **118** (2022), 577–586, doi:10.1007/s00013-022-01724-2, https://doi.org/10.1007/s00013-022-01724-2.

[6] R. Frucht and F. Harary, On the corona of two graphs, *Aequationes Math.* **4** (1970), 322–325, doi:10.1007/BF01844162, https://doi.org/10.1007/BF01844162.

[7] I. Gutman, Independent vertex sets in some compound graphs, *Publ. Inst. Math. (Beograd) (N.S.)* **52** (1992), 5–9.

[8] I. Gutman and F. Harary, Generalizations of the matching polynomial, *Util. Math.* **24** (1983), 97–106.

[9] C. Hoede and X. Li, Clique polynomials and independent set polynomials of graphs, *Discrete Math.* **125** (1994), 219–228, doi:10.1016/0012-365X(94)90163-5, https://doi.org/10.1016/0012-365X(94)90163-5.

[10] V. E. Levit and E. Mandrescu, A family of well-covered graphs with unimodal independence polynomials, *Congr. Numer.* **165** (2003), 195–207.

[11] V. E. Levit and E. Mandrescu, On unimodality of independence polynomials of some well-covered trees, *Discrete Math. Theor. Comput. Sci.* **2731** (2003), 237–256.

[12] V. E. Levit and E. Mandrescu, Independence polynomials and the unimodality conjecture for very well-covered, quasi-regularizable, and perfect graphs, in: *Graph Theory Trends in Mathematics*, Birkhäuser, Basel, pp. 243–254, 2006.

[13] V. E. Levit and E. Mandrescu, On the roots of independence polynomials of almost all very well-covered graphs, *Discrete Appl. Math.* **156** (2008), 478–491, doi:10.1016/j.dam.2006.06.016, https://doi.org/10.1016/j.dam.2006.06.016.

[14] V. E. Levit and E. Mandrescu, On the independence polynomial of the corona of graphs, *Discrete Appl. Math.* **203** (2016), 85–93, doi:10.1016/j.dam.2015.09.021, https://doi.org/10.1016/j.dam.2015.09.021.

[15] V. E. Levit and E. Mandrescu, On clique corona graphs–a short survey, *Mat. Contemp.* **45** (2017), 68–76.

[16] V. E. Levit and E. Mandrescu, The roller-coaster conjecture revisited, *Graphs Comb.* **33** (2017), 1499–1508, doi:10.1007/s00373-017-1834-z, https://doi.org/10.1007/s00373-017-1834-z.

[17] V. E. Levit and E. Mandrescu, Graph operations preserving $\mathbf{W}_2$-property, *Electron. Notes Discrete Math.* **68** (2018), 35–40, doi:10.1016/j.endm.2018.06.007, https://doi.org/10.1016/j.endm.2018.06.007.

[18] V. E. Levit and E. Mandrescu, 1-well-covered graphs revisited, *Eur. J. Comb.* **80** (2019), 261–272, doi:10.1016/j.ejc.2018.02.021, https://doi.org/10.1016/j.ejc.2018.02.021.

[19] T. Michael and W. Traves, Independence sequences of well-covered graphs: non-unimodality and the roller-coaster conjecture, *Graphs Comb.* **19** (2003), 403–411, doi:10.1007/s00373-002-0515-7, https://doi.org/10.1007/s00373-002-0515-7.

[20] M. R. Pinter, $\mathbf{W}_2$ *graphs and strongly well-covered graphs: two well-covered graph subclasses*, Ph.D. thesis, Vanderbilt University, 1991.

Accepted manuscript

[21] M. D. Plummer, Some covering concepts in graphs, *J. Comb. Theory* **8** (1970), 91–98, doi: 10.1016/S0021-9800(70)80011-4, https://doi.org/10.1016/S0021-9800(70)80011-4.

[22] R. P. Stanley, Log-concave and unimodal sequences in algebra, combinatorics, and geometry, *Ann. N. Y. Acad. Sci.* **576** (1989), 500–535, doi:10.1111/j.1749-6632.1989.tb16434.x, https://doi.org/10.1111/j.1749-6632.1989.tb16434.x.

[23] J. A. W. Staples, On some subclasses of well-covered graphs, *J. Graph Theory* **3** (1979), 197–204, doi:10.1002/jgt.3190030211, https://doi.org/10.1002/jgt.3190030211.

[24] J. W. Staples, *On some subclasses of well-covered graphs*, Ph.D. thesis, Vanderbilt University, 1975.

[25] J. Topp and L. Volkmann, On the well-coveredness of products of graphs, *Ars Comb.* **33** (1992), 199–215.

Accepted manuscript